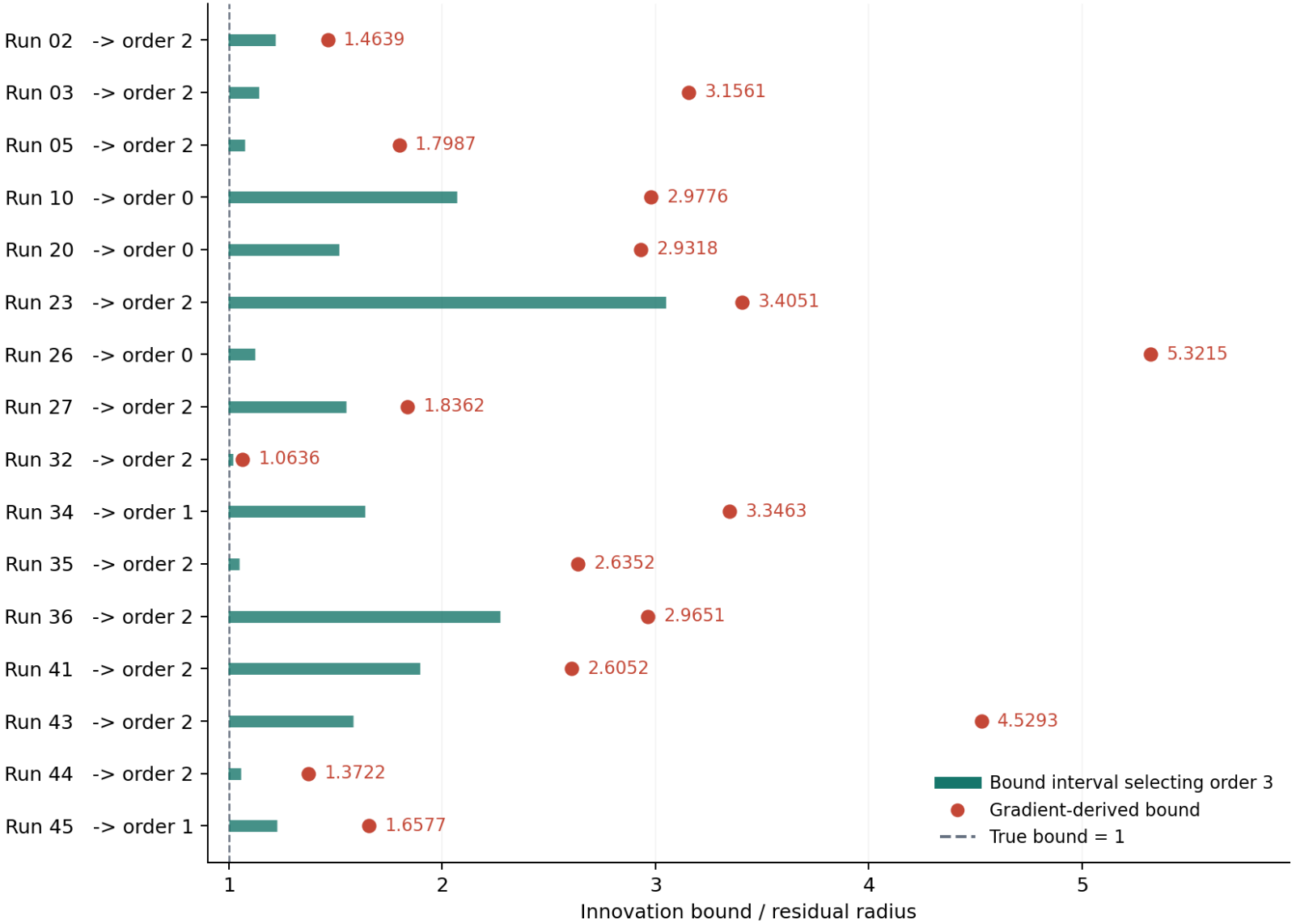
All 16 additional underselections

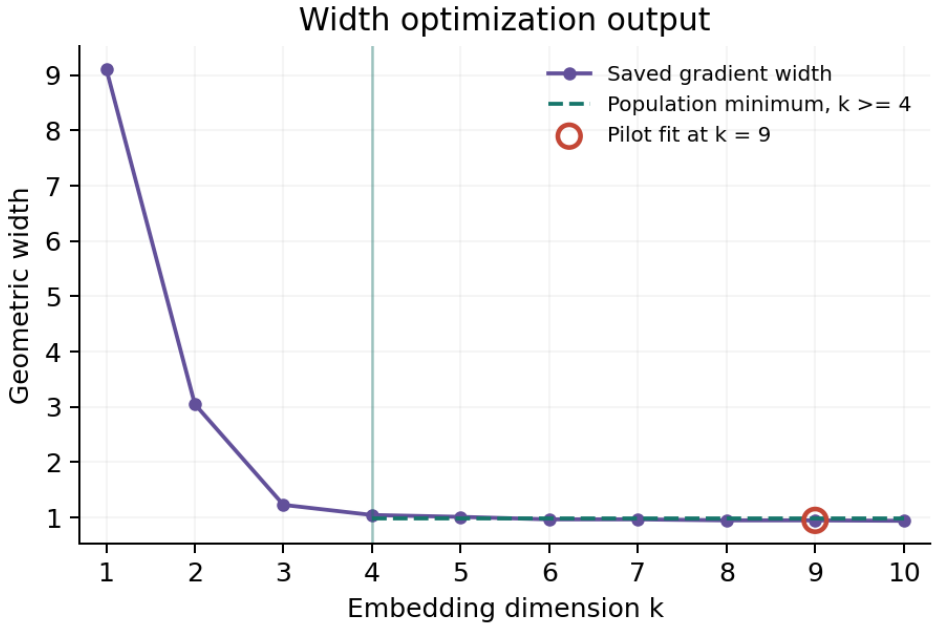
True order = 3; true bound = 1. Each red estimate exceeds the interval selecting order 3.

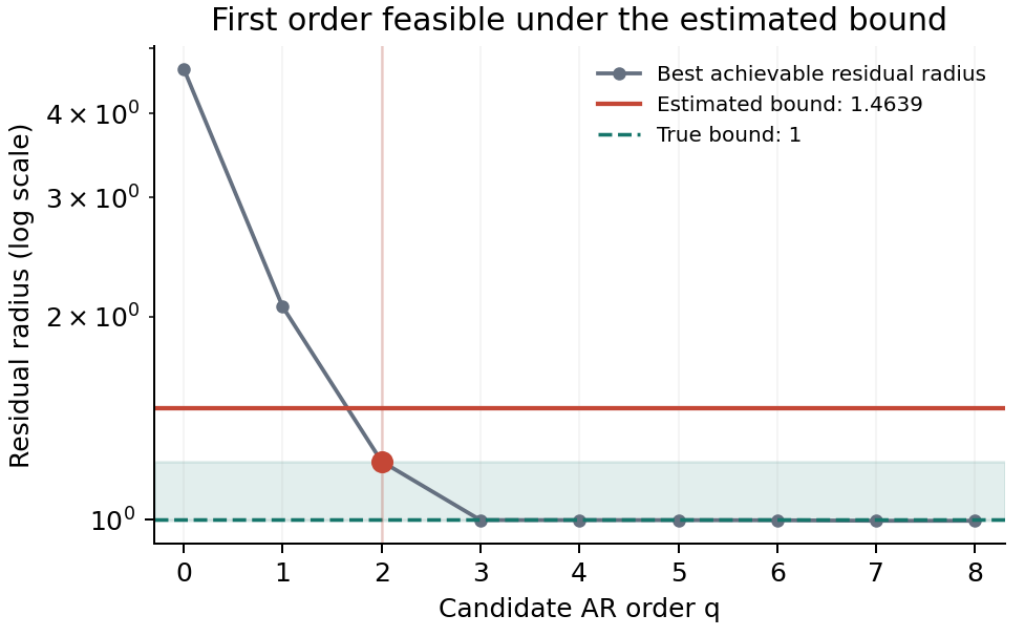


Saved padded-start fits at embedding k = 9 (8 lags), n = 1992. Known-bound selection returns order 3 in every case.
Original diagnostic calculation; all 16 cases are included. No gradient fits were rerun.

Run 02 | True order 3 | Selected order 2

Bound calculation: 0.506673574 / 0.346119139 = 1.463870436 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998674569 <= bound < 1.219936966. Estimated bound 1.463870436 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.5416771, -0.8888877, 0.1631933)

Original width: 0.947025881; corrected projection width: 0.987810977; maximum absolute residual at t=282.

Original direction u = (-0.01300, 0.02440, -0.08236, 0.46594, -0.70411, 0.49929, -0.17300, 0.01930, 0.00000)

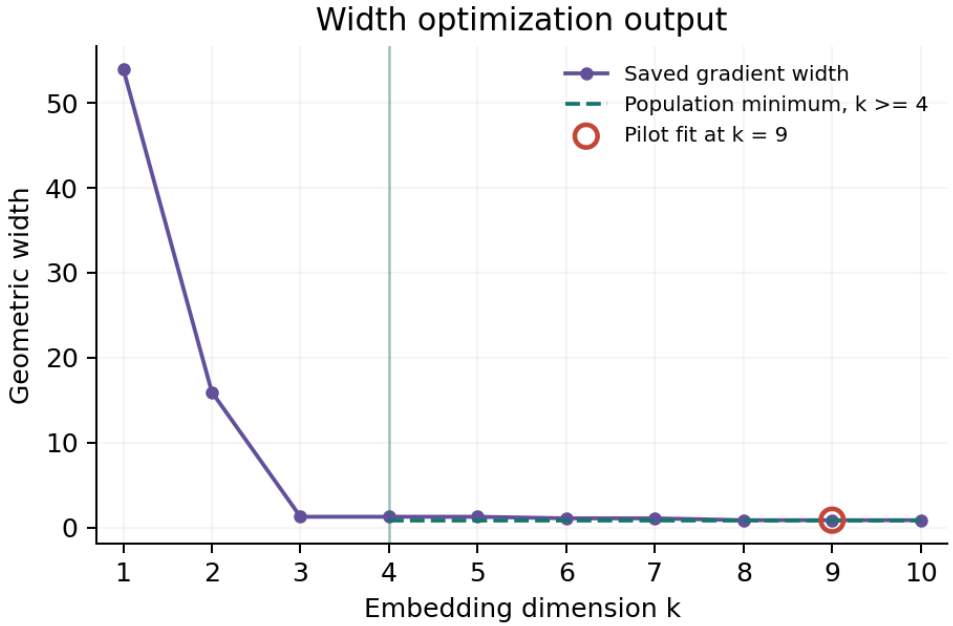
Corrected direction v = (0.34612, -0.68064, 0.58158, -0.26942, 0.07480, -0.02168, 0.00643, -0.00072, 0.00000)

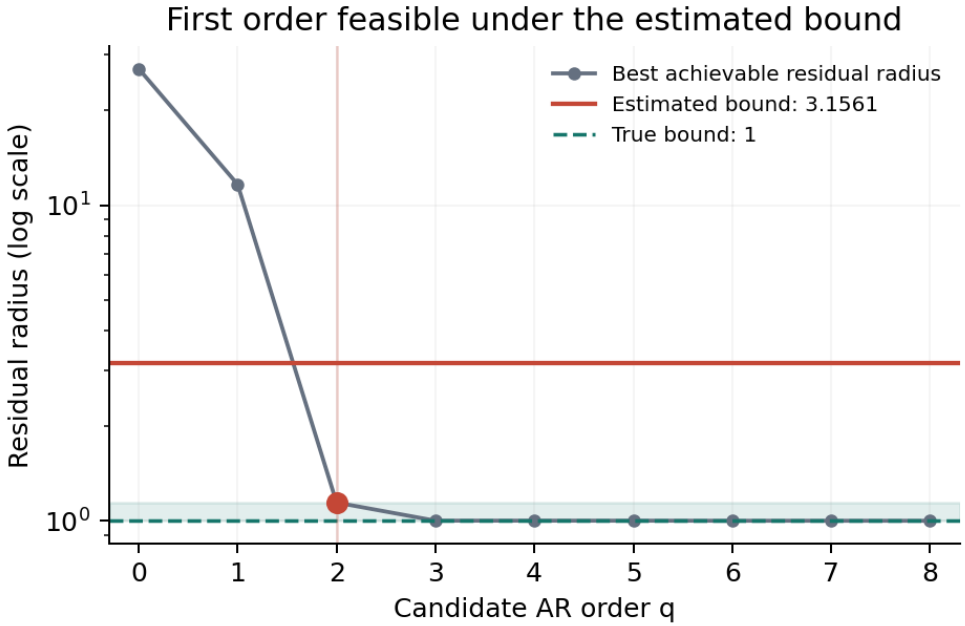
Inputs: ar3_order_selection_warm_starts/processes/process_002.csv and replications/replication_002.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 03 | True order 3 | Selected order 2

Bound calculation: 0.444238630 / 0.140753709 = 3.156141563 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.999241792 <= bound < 1.140729366. Estimated bound 3.156141563 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.9840462, -1.2884123, 0.1631563)

Original width: 0.849818271; corrected projection width: 0.884002274; maximum absolute residual at t=1725.

Original direction u = (-0.09563, 0.29186, -0.49648, 0.60235, -0.47819, 0.23734, -0.10040, 0.03569, 0.00000)

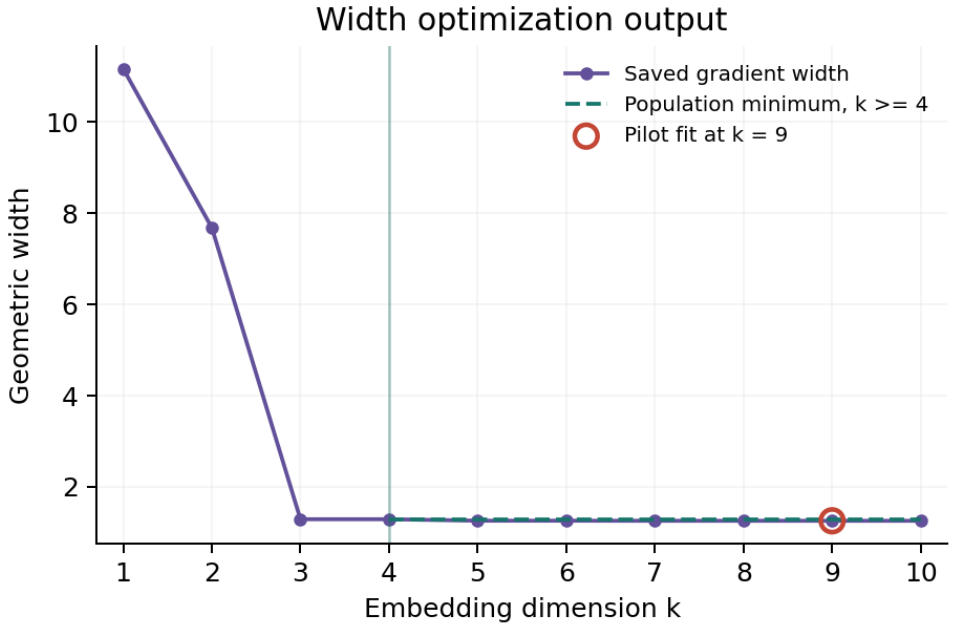
Corrected direction v = (0.14075, -0.40969, 0.56723, -0.54423, 0.39107, -0.18909, 0.07164, -0.02425, -0.00000)

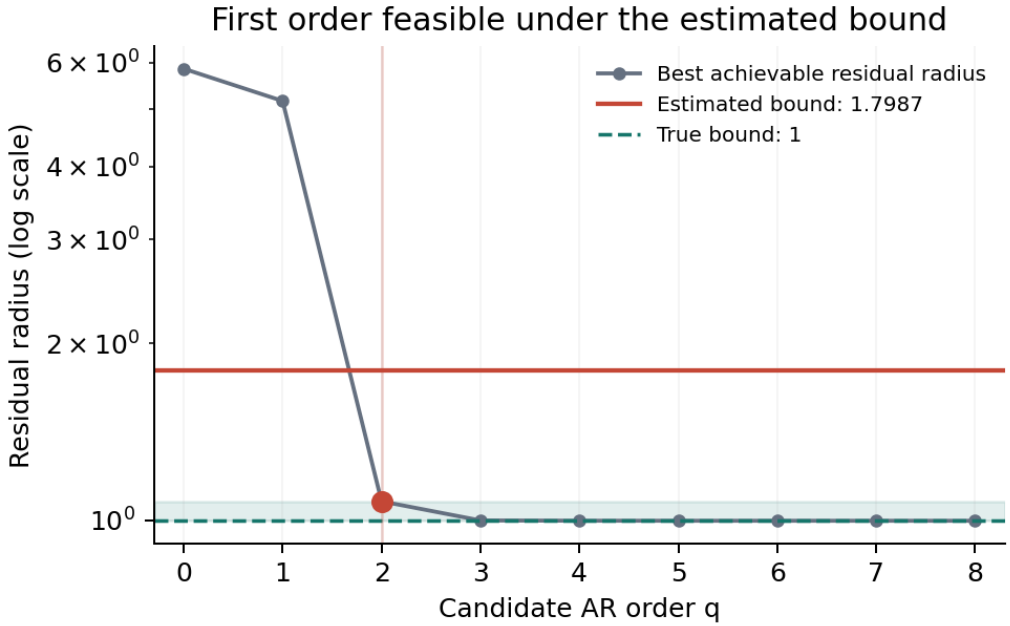
Inputs: ar3_order_selection_warm_starts/processes/process_003.csv and replications/replication_003.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 05 | True order 3 | Selected order 2

Bound calculation: 0.691885497 / 0.384648697 = 1.798746501 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998780705 <= bound < 1.076471751. Estimated bound 1.798746501 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (0.8522412, -0.8511510, -0.0826316)

Original width: 1.256417937; corrected projection width: 1.348532207; maximum absolute residual at t=508.

Original direction u = (0.00065, 0.00991, -0.12033, 0.53619, -0.63513, 0.52625, -0.13263, 0.00000, 0.00000)

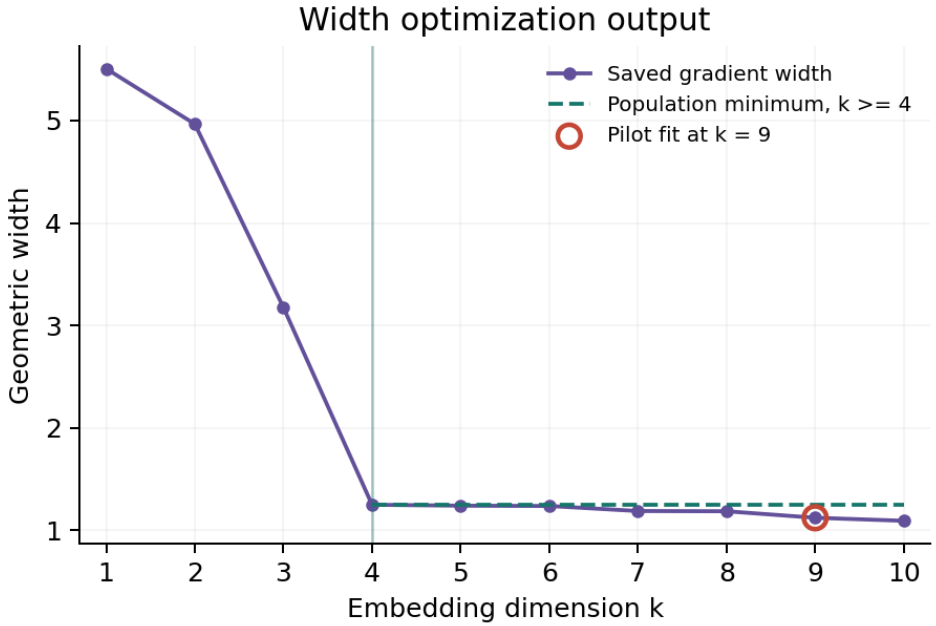
Corrected direction v = (0.38465, -0.60224, 0.63808, -0.28232, 0.04984, -0.00288, -0.00022, 0.00000, 0.00000)

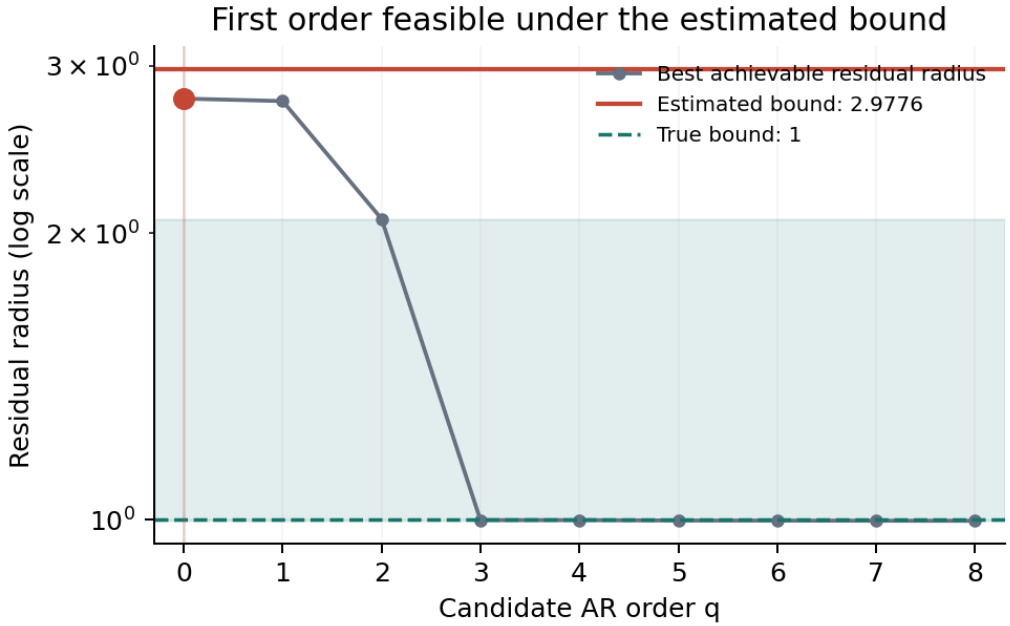
Inputs: ar3_order_selection_warm_starts/processes/process_005.csv and replications/replication_005.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 10 | True order 3 | Selected order 0

Bound calculation: 0.677557114 / 0.227552102 = 2.977591102 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998752109 <= bound < 2.070681416. Estimated bound 2.977591102 admits order 0.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (0.6976038, -0.8139940, 0.6426158)

Original width: 1.120481298; corrected projection width: 1.268954703; maximum absolute residual at t=1809.

Original direction u = (0.12033, -0.23689, 0.44591, -0.43757, 0.47122, -0.41622, 0.28409, -0.21494, 0.13001)

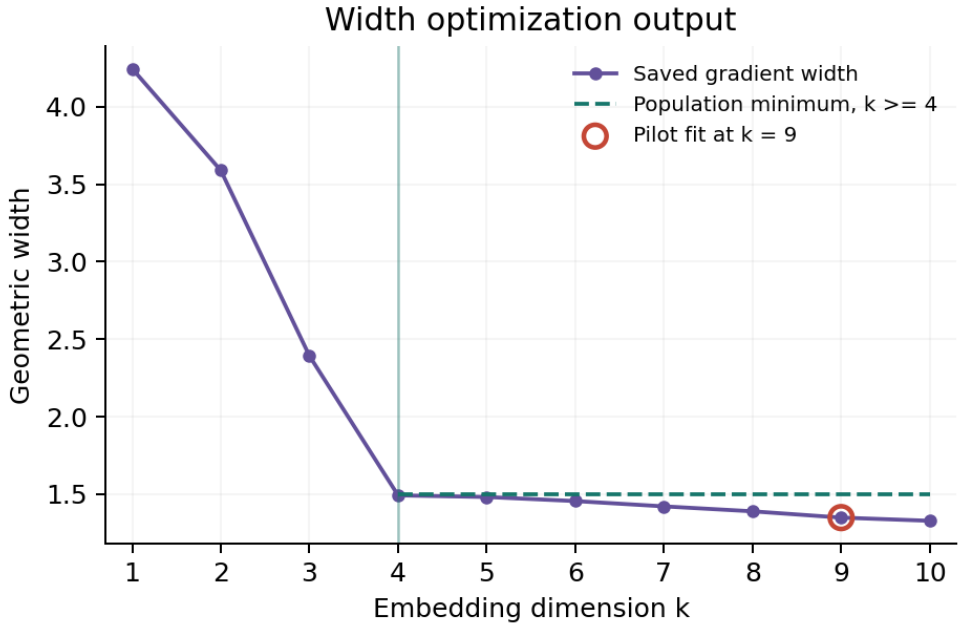
Corrected direction v = (0.22755, -0.30521, 0.44229, -0.49200, 0.42815, -0.35161, 0.28482, -0.15680, 0.06875)

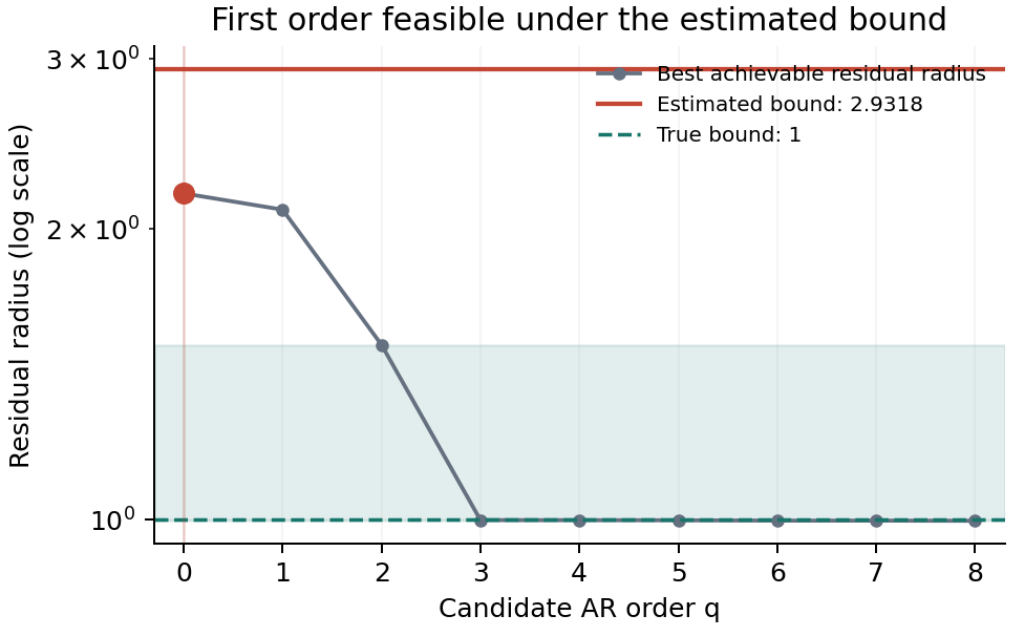
Inputs: ar3_order_selection_warm_starts/processes/process_010.csv and replications/replication_010.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 20 | True order 3 | Selected order 0

Bound calculation: 0.738180799 / 0.251781786 = 2.931827644 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998883078 <= bound < 1.517404614. Estimated bound 2.931827644 admits order 0.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (0.5198639, -0.6146464, 0.3588801)

Original width: 1.348659322; corrected projection width: 1.466869321; maximum absolute residual at t=882.

Original direction u = (-0.05580, 0.15290, -0.32365, 0.56450, -0.51774, 0.43895, -0.26982, 0.11282, -0.06195)

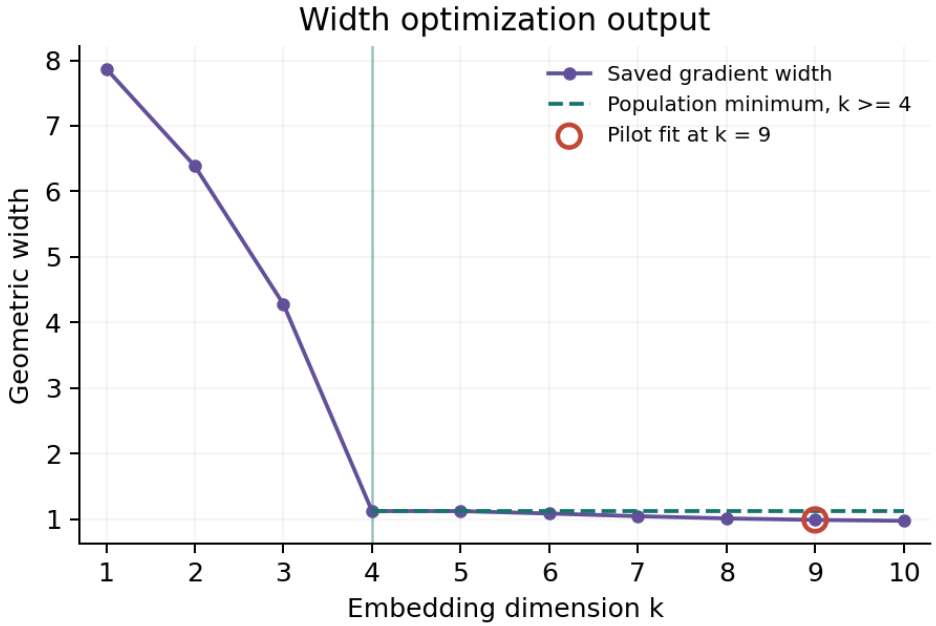
Corrected direction v = (0.25178, -0.41604, 0.52866, -0.52747, 0.36187, -0.24551, 0.11312, -0.03994, 0.01373)

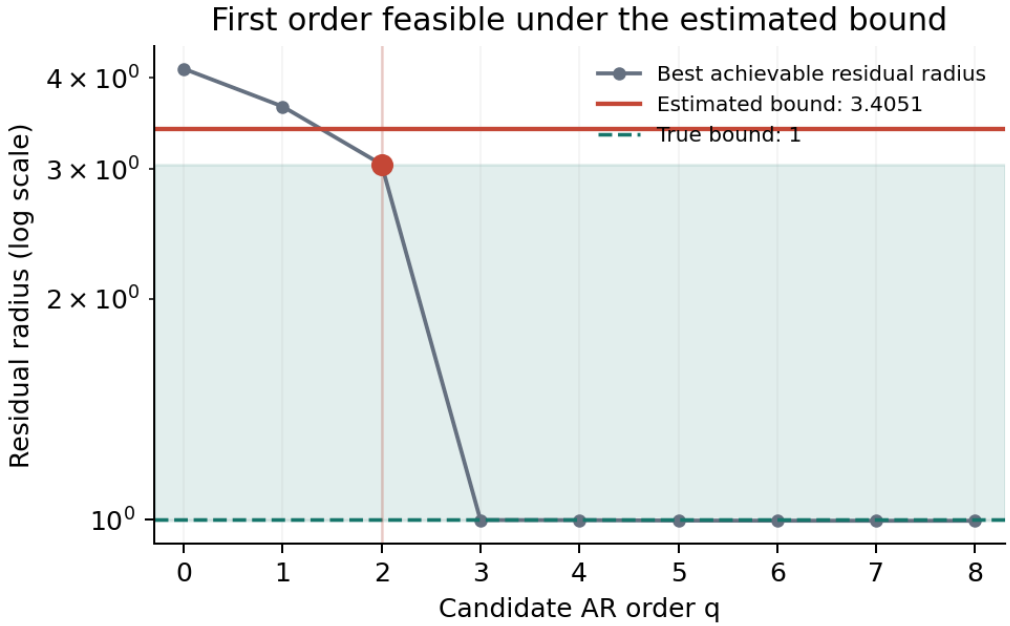
Inputs: ar3_order_selection_warm_starts/processes/process_020.csv and replications/replication_020.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 23 | True order 3 | Selected order 2

Bound calculation: 0.568321317 / 0.166902023 = 3.405119405 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998672912 <= bound < 3.046995482. Estimated bound 3.405119405 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-0.8815031, -0.8612243, -0.8236825)

Original width: 0.986444274; corrected projection width: 1.103918040; maximum absolute residual at t=1214.

Original direction u = (0.06526, 0.13586, 0.35614, 0.48392, 0.53054, 0.48838, 0.27756, 0.13539, 0.03013)

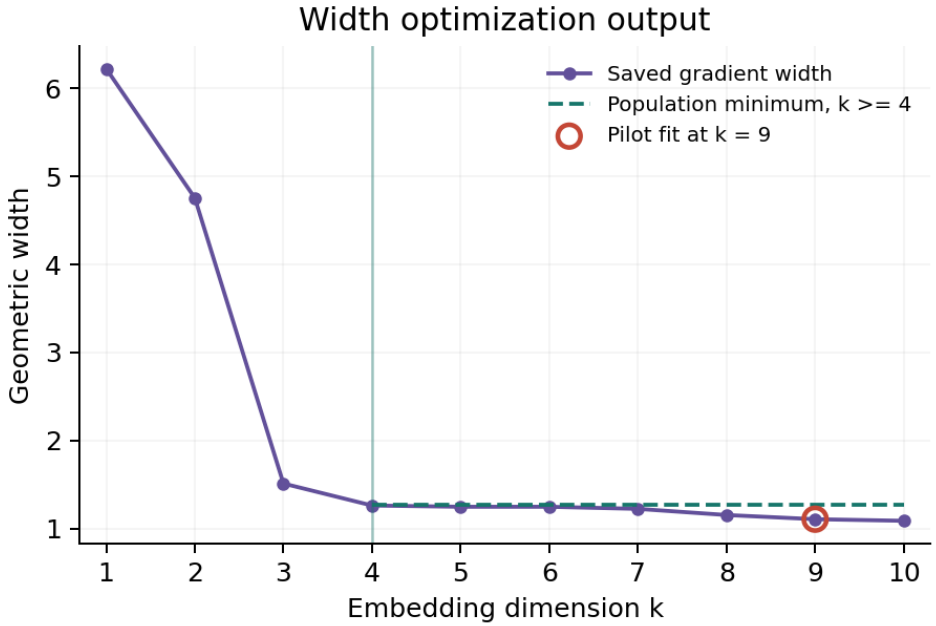
Corrected direction v = (0.16690, 0.29924, 0.47874, 0.56183, 0.45399, 0.32614, 0.14822, 0.05634, 0.01178)

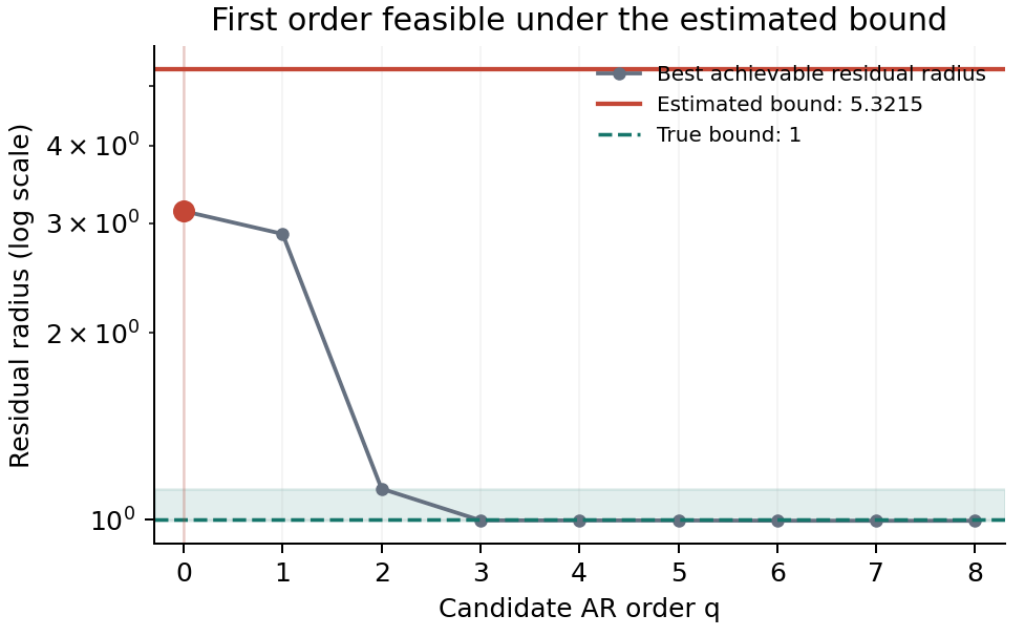
Inputs: ar3_order_selection_warm_starts/processes/process_023.csv and replications/replication_023.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 26 | True order 3 | Selected order 0

Bound calculation: 0.593030698 / 0.111439872 = 5.321530679 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.997969347 <= bound < 1.122240185. Estimated bound 5.321530679 admits order 0.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-0.8296310, -0.8968777, -0.1177140)

Original width: 1.104511786; corrected projection width: 1.185615766; maximum absolute residual at t=1411.

Original direction u = (-0.06817, -0.13642, -0.28260, -0.42342, -0.52671, -0.49554, -0.38255, -0.20436, -0.08062)

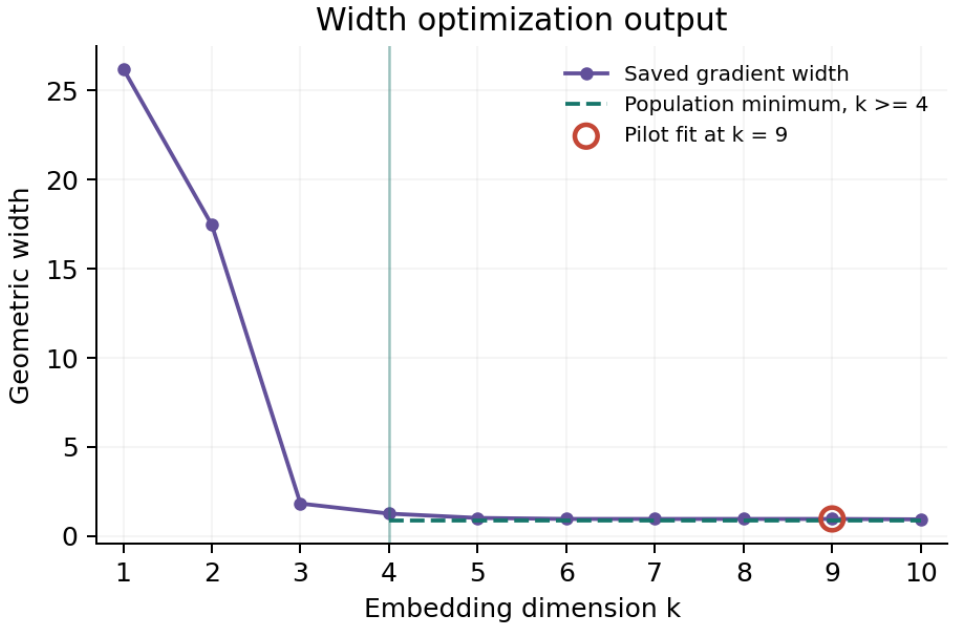
Corrected direction v = (0.11144, 0.25201, 0.42547, 0.51241, 0.50777, 0.38314, 0.24667, 0.11218, 0.04932)

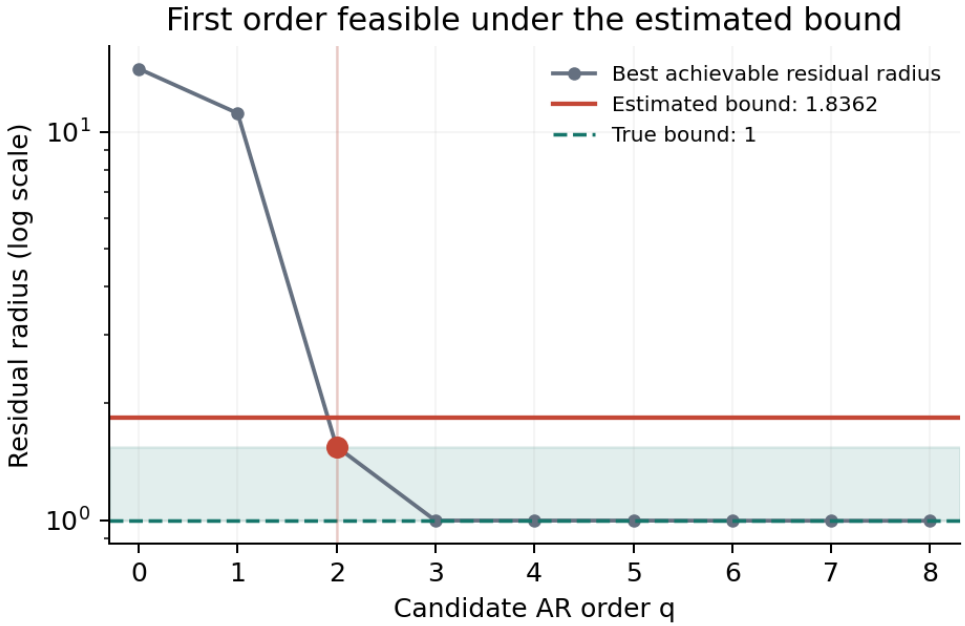
Inputs: ar3_order_selection_warm_starts/processes/process_026.csv and replications/replication_026.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 27 | True order 3 | Selected order 2

Bound calculation: 0.466153561 / 0.253865681 = 1.836221261 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.999155028 <= bound < 1.548094462. Estimated bound 1.836221261 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-1.4630242, -1.4461715, -0.4546358)

Original width: 0.961966195; corrected projection width: 0.925814077; maximum absolute residual at t=1321.

Original direction u = (0.17322, 0.30416, 0.55406, 0.57823, 0.44497, 0.19531, 0.00000, 0.00000, 0.00000)

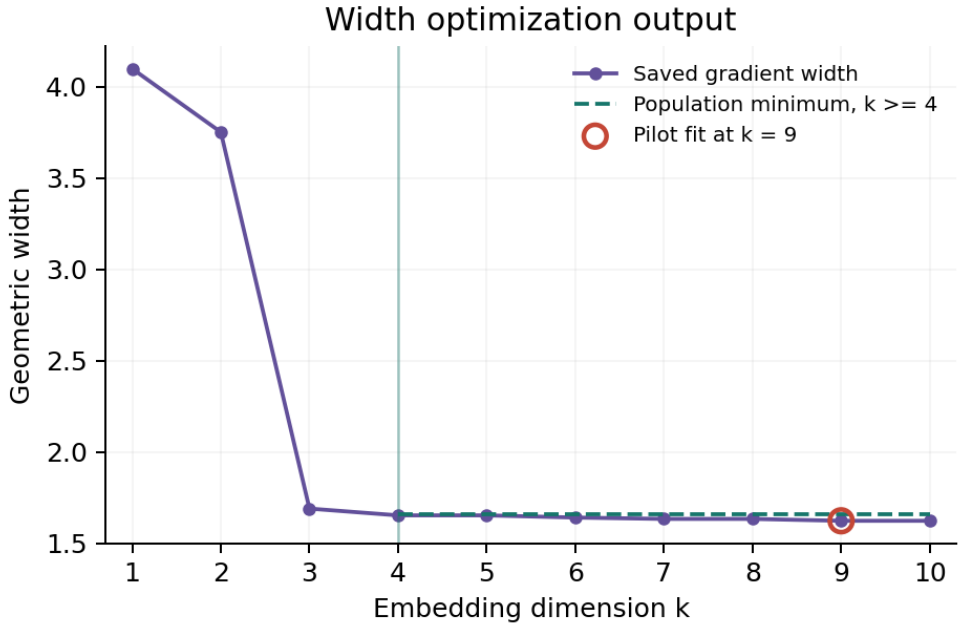
Corrected direction v = (0.25387, 0.44705, 0.61544, 0.49739, 0.30294, 0.13327, 0.00000, 0.00000, 0.00000)

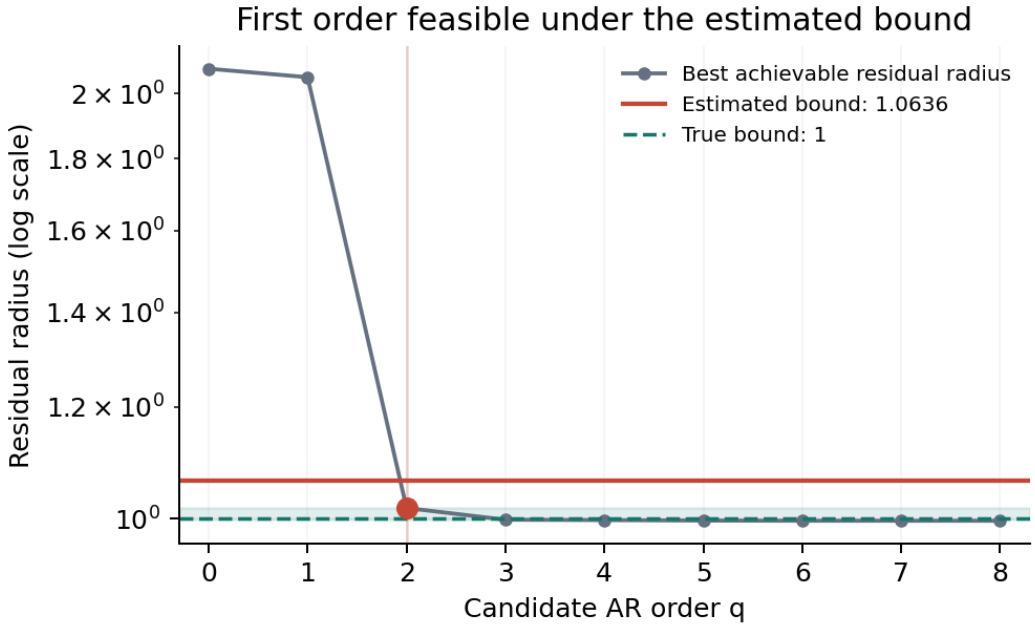
Inputs: ar3_order_selection_warm_starts/processes/process_027.csv and replications/replication_027.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 32 | True order 3 | Selected order 2

Bound calculation: 0.840235625 / 0.790004673 = 1.063583107 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998576621 <= bound < 1.017791572. Estimated bound 1.063583107 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-0.2889160, -0.6100022, -0.0288156)

Original width: 1.622468190; corrected projection width: 1.678137319; maximum absolute residual at t=1687.

Original direction u = (-0.01001, -0.01382, -0.05574, -0.03378, -0.82364, -0.25245, -0.50204, -0.03408, -0.01172)

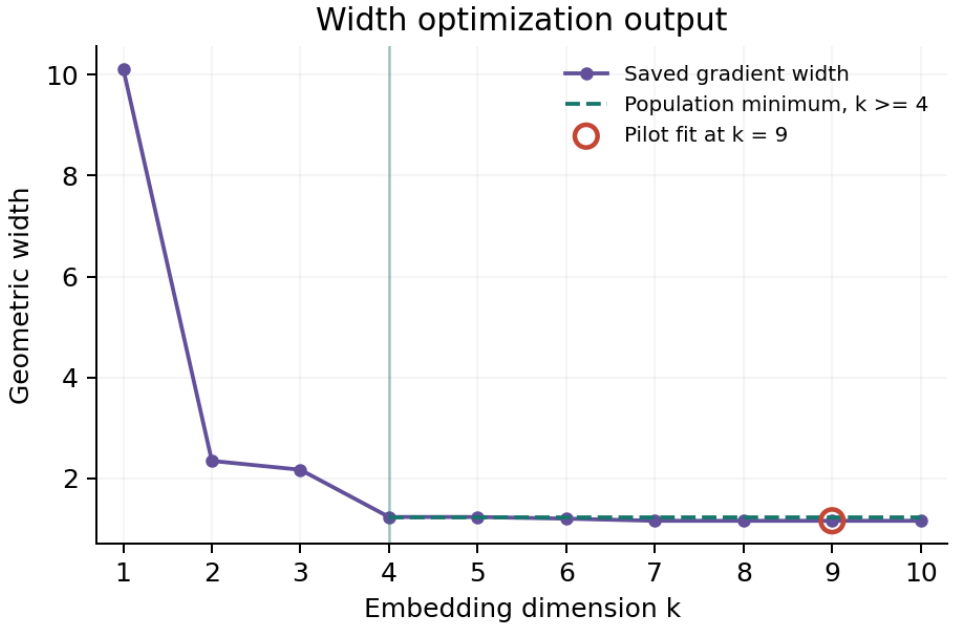
Corrected direction v = (0.79000, 0.25485, 0.55073, 0.06665, 0.05478, 0.01205, 0.00749, 0.00059, 0.00015)

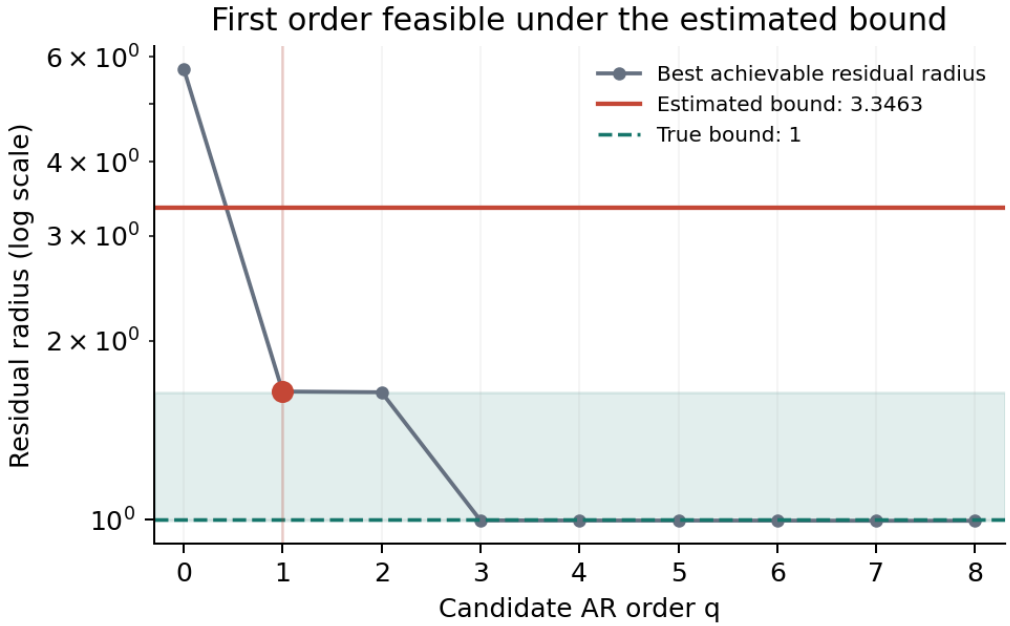
Inputs: ar3_order_selection_warm_starts/processes/process_032.csv and replications/replication_032.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 34 | True order 3 | Selected order 1

Bound calculation: 0.619651899 / 0.185176471 = 3.346277719 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.997518657 <= bound < 1.638396553. Estimated bound 3.346277719 admits order 1.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.0816925, -0.5696289, 0.4494789)

Original width: 1.155779386; corrected projection width: 1.212839833; maximum absolute residual at t=893.

Original direction u = (0.06455, -0.22494, 0.46499, -0.60787, 0.49398, -0.31371, 0.13072, 0.00000, 0.00000)

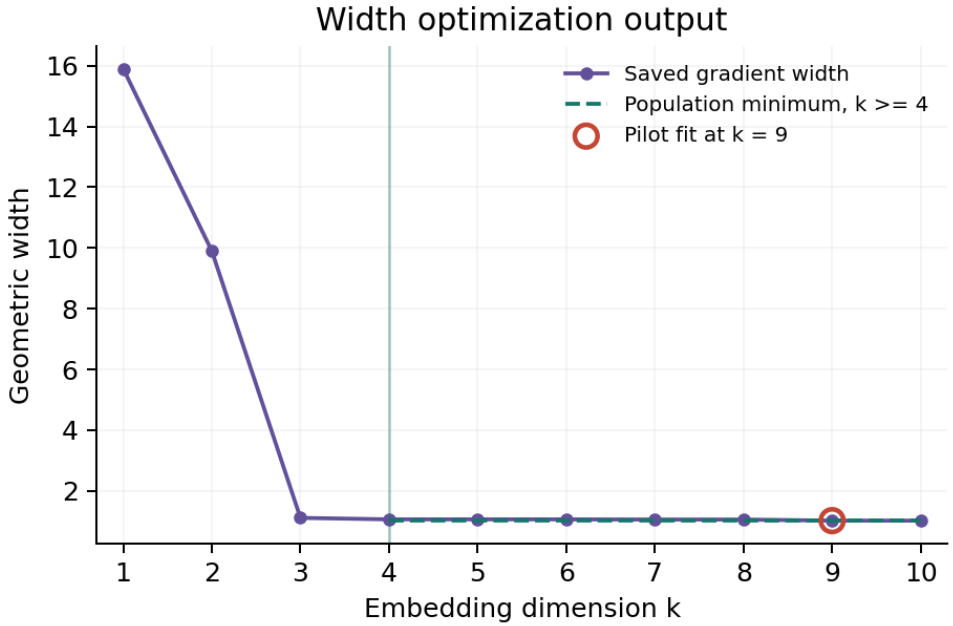
Corrected direction v = (0.18518, -0.44084, 0.55985, -0.54602, 0.36364, -0.15967, 0.04557, 0.00000, 0.00000)

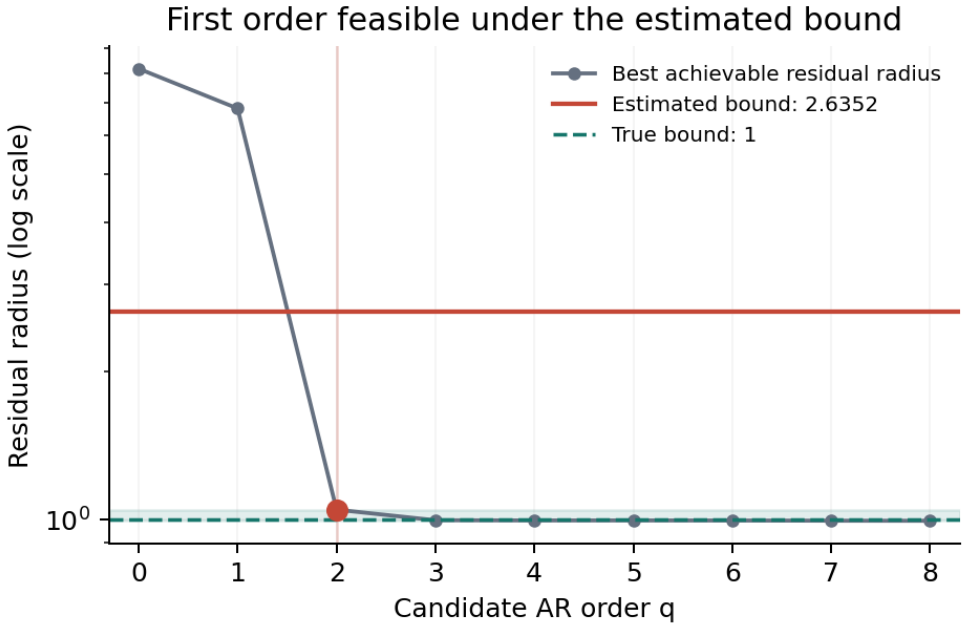
Inputs: ar3_order_selection_warm_starts/processes/process_034.csv and replications/replication_034.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 35 | True order 3 | Selected order 2

Bound calculation: 0.546108000 / 0.207237949 = 2.635173733 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.997111343 <= bound < 1.047581258. Estimated bound 2.635173733 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.3533191, -1.0329073, 0.0599311)

Original width: 1.017565902; corrected projection width: 1.079395540; maximum absolute residual at t=1692.

Original direction u = (-0.14833, 0.45168, -0.59535, 0.46165, -0.24404, 0.19329, -0.24421, 0.20077, -0.09778)

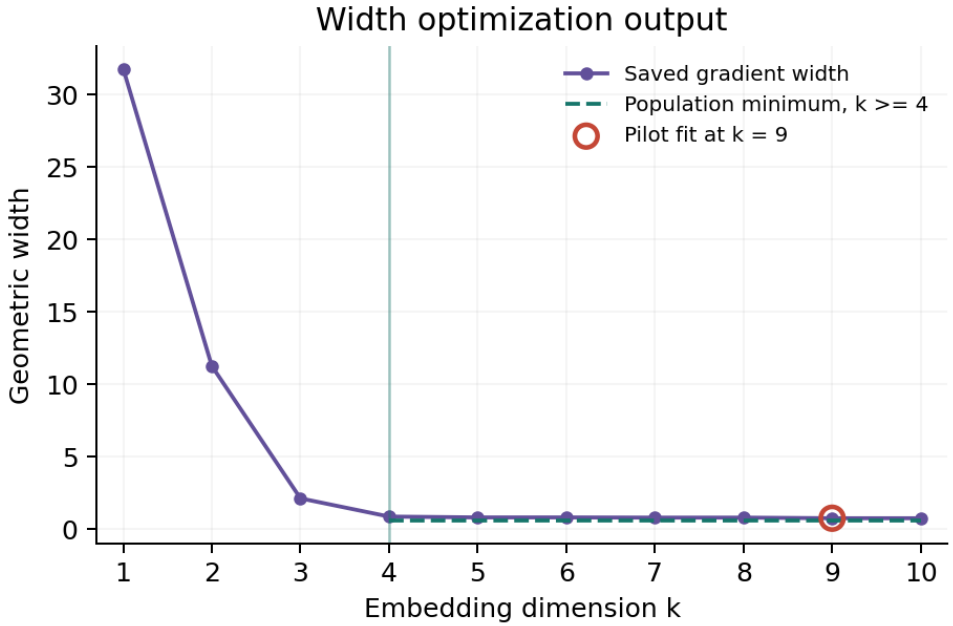
Corrected direction v = (0.20724, -0.49399, 0.59213, -0.41398, 0.21992, -0.20942, 0.24044, -0.18999, 0.06998)

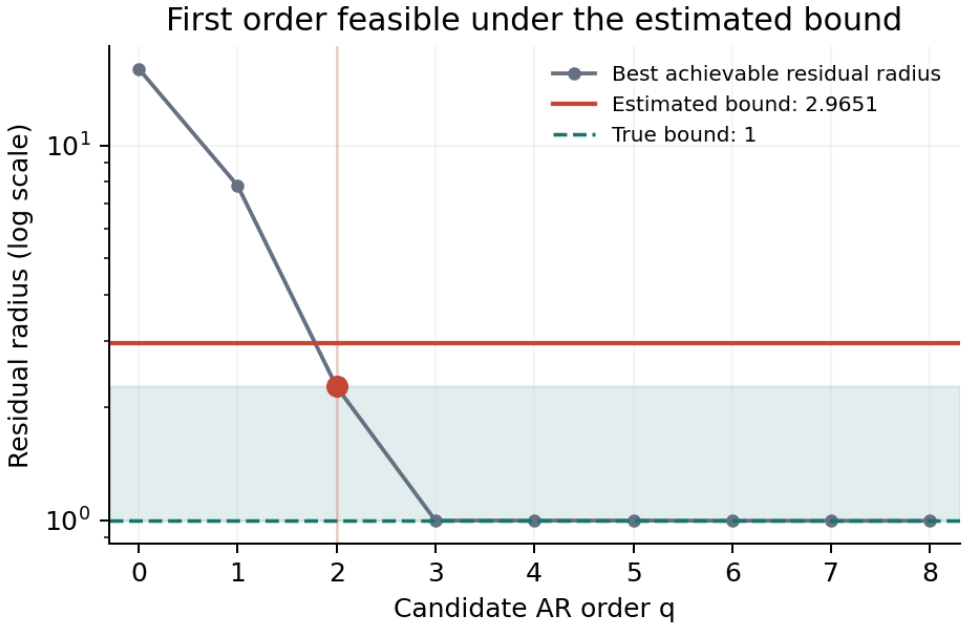
Inputs: ar3_order_selection_warm_starts/processes/process_035.csv and replications/replication_035.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 36 | True order 3 | Selected order 2

Bound calculation: 0.366399774 / 0.123568828 = 2.965147270 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998337848 <= bound < 2.273337743. Estimated bound 2.965147270 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-2.3888767, -2.1701322, -0.7187614)

Original width: 0.756715141; corrected projection width: 0.708334987; maximum absolute residual at t=610.

Original direction u = (-0.05510, -0.22560, -0.43703, -0.53792, -0.50635, -0.39786, -0.22226, -0.03504, 0.02004)

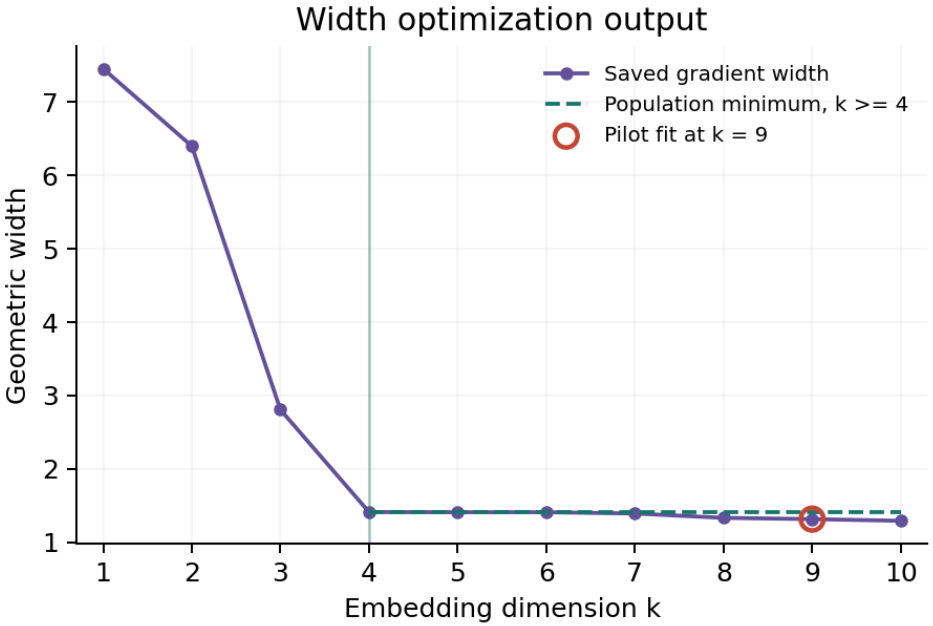
Corrected direction v = (0.12357, 0.34538, 0.49252, 0.53068, 0.47694, 0.31635, 0.11660, 0.00401, -0.00894)

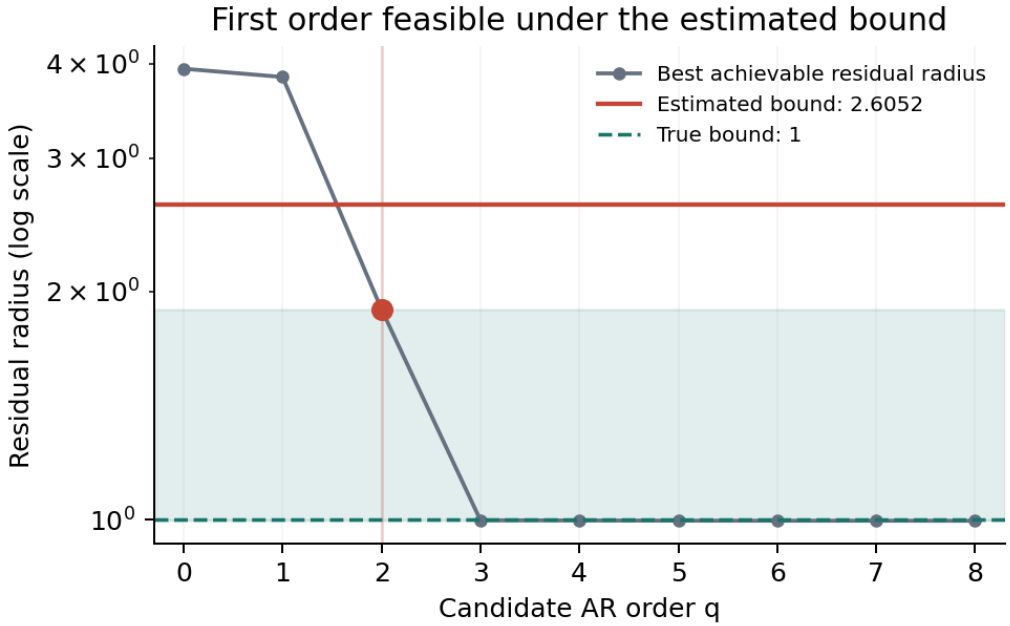
Inputs: ar3_order_selection_warm_starts/processes/process_036.csv and replications/replication_036.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 41 | True order 3 | Selected order 2

Bound calculation: 0.702787686 / 0.269768523 = 2.605150809 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.998688245 <= bound < 1.895442551. Estimated bound 2.605150809 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (0.4025869, -0.7018632, 0.5975948)

Original width: 1.315924683; corrected projection width: 1.399282972; maximum absolute residual at t=1936.

Original direction u = (0.07892, -0.27810, 0.25267, -0.56001, 0.45883, -0.40996, 0.35581, -0.15202, 0.10330)

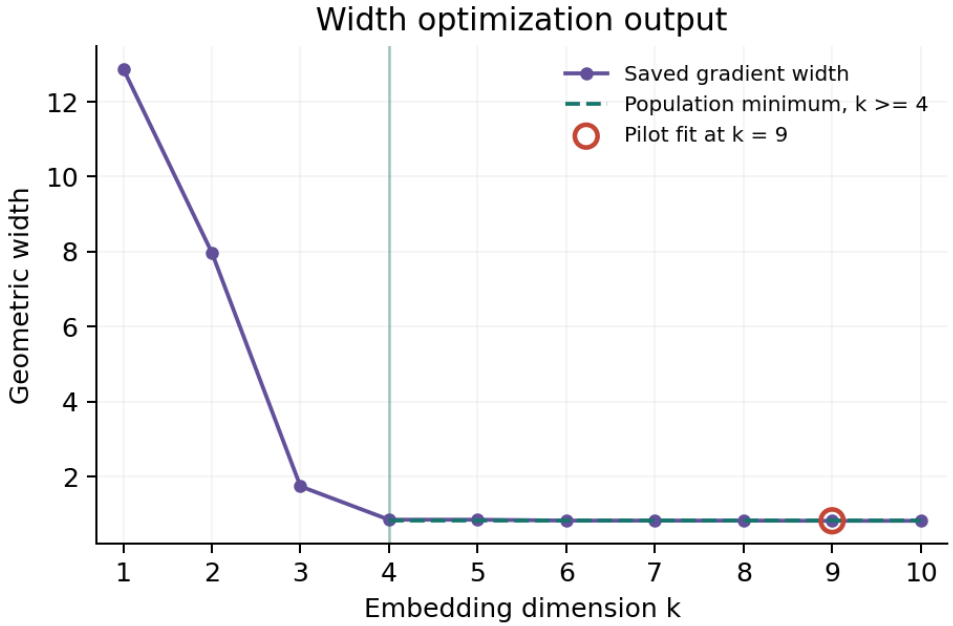
Corrected direction v = (0.26977, -0.21397, 0.48035, -0.50920, 0.41673, -0.39936, 0.20302, -0.12700, 0.03022)

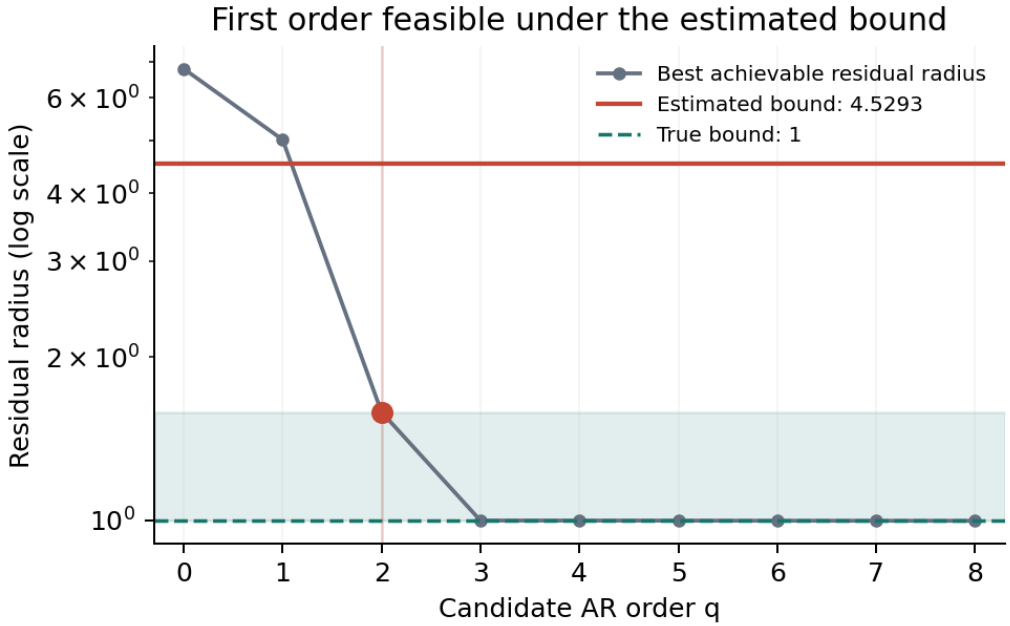
Inputs: ar3_order_selection_warm_starts/processes/process_041.csv and replications/replication_041.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 43 | True order 3 | Selected order 2

Bound calculation: 0.422339099 / 0.093246192 = 4.529290575 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.999483867 <= bound < 1.582847503. Estimated bound 4.529290575 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (-1.5984070, -1.4013179, -0.4074759)

Original width: 0.822675112; corrected projection width: 0.822709549; maximum absolute residual at t=764.

Original direction u = (-0.04933, -0.17791, -0.38228, -0.54069, -0.54599, -0.40450, -0.23231, -0.10533, -0.02535)

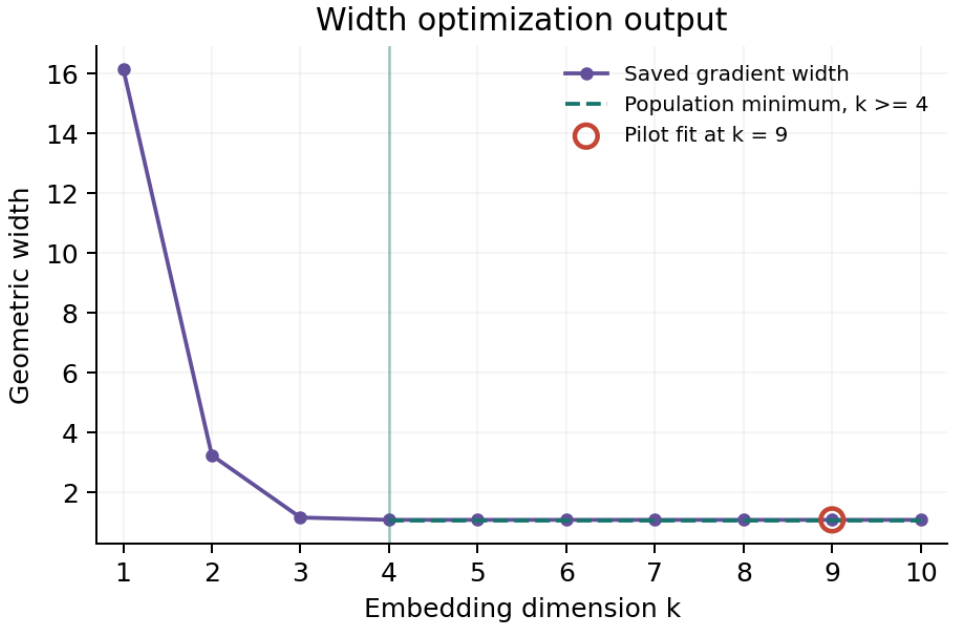
Corrected direction v = (0.09325, 0.28832, 0.48494, 0.55538, 0.47925, 0.32211, 0.16441, 0.06262, 0.01341)

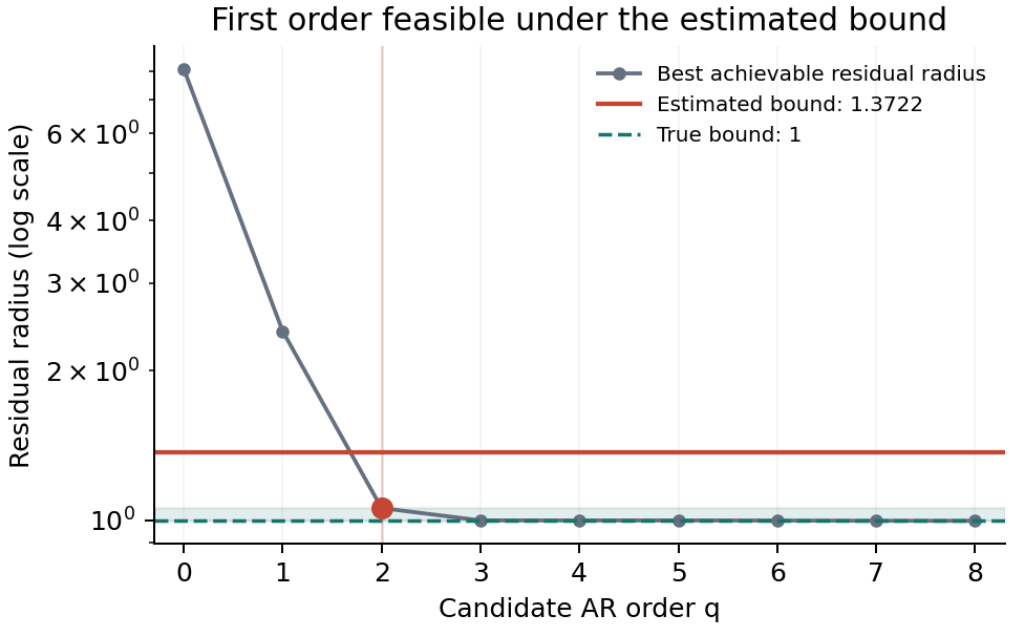
Inputs: ar3_order_selection_warm_starts/processes/process_043.csv and replications/replication_043.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 44 | True order 3 | Selected order 2

Bound calculation: 0.575984142 / 0.419741798 = 1.372234418 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.999140400 <= bound < 1.057945891. Estimated bound 1.372234418 admits order 2.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.5212923, -0.5325149, -0.0697649)

Original width: 1.084638194; corrected projection width: 1.138915617; maximum absolute residual at t=1232.

Original direction u = (-0.09907, 0.58142, -0.75353, 0.29040, 0.00000, 0.00000, 0.00000, 0.00000, 0.00000)

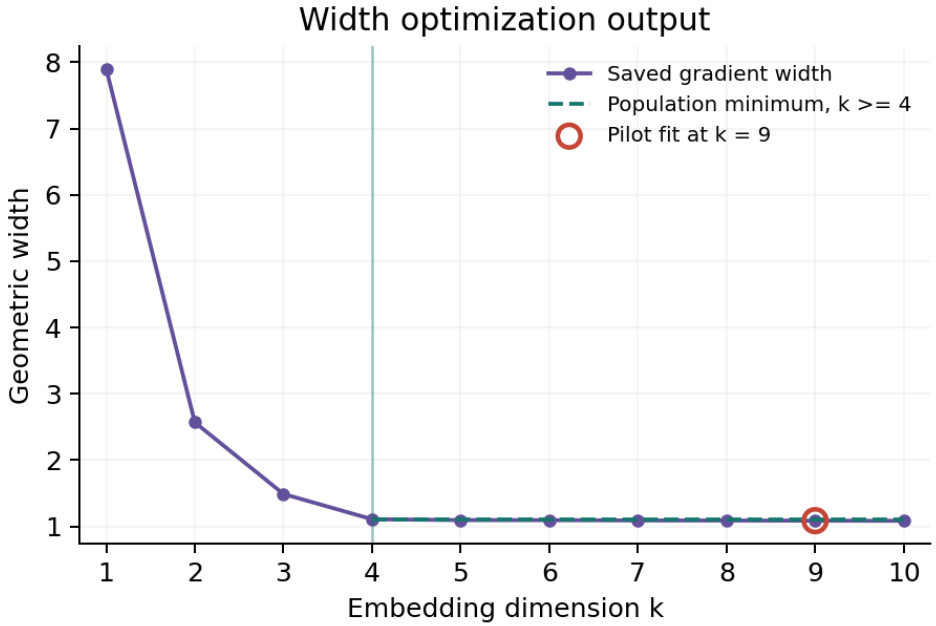
Corrected direction v = (0.41974, -0.78406, 0.45208, -0.06854, 0.00000, 0.00000, 0.00000, 0.00000, 0.00000)

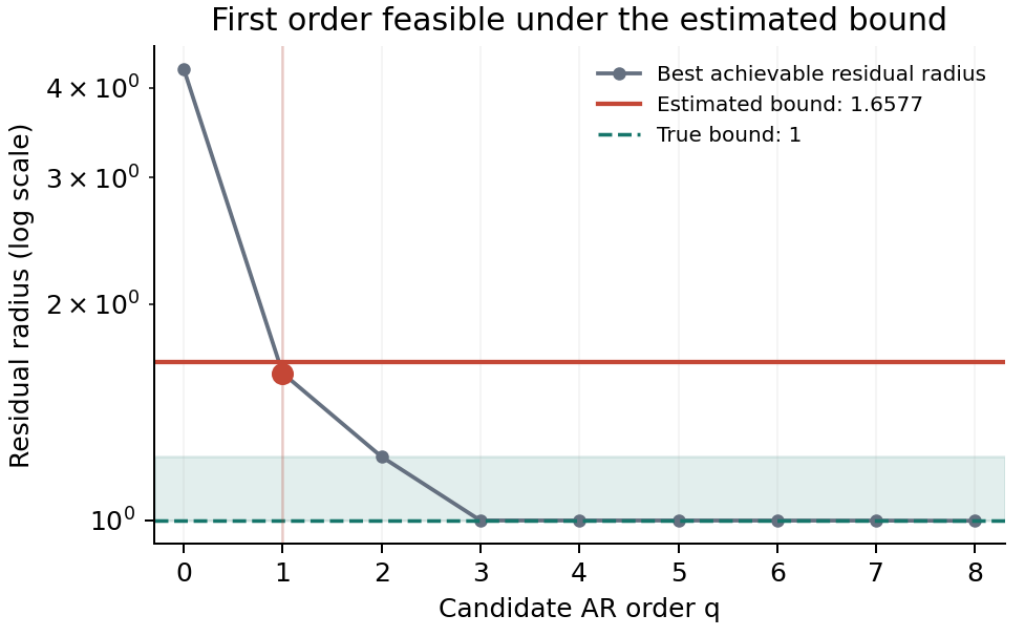
Inputs: ar3_order_selection_warm_starts/processes/process_044.csv and replications/replication_044.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.

Run 45 | True order 3 | Selected order 1

Bound calculation: 0.586187982 / 0.353624047 = 1.657658712 (maximum absolute corrected projection / first coordinate)





Order 3 requires 0.999272696 <= bound < 1.225849573. Estimated bound 1.657658712 admits order 1.

Left: saved per-dimension observation windows. Right: common t = 9,...,2000. Shading marks the interval selecting order 3.

True AR coefficients: (1.3074961, -0.7108263, 0.2007344)

Original width: 1.087072008; corrected projection width: 1.160066088; maximum absolute residual at t=1816.

Original direction u = (-0.01471, 0.01223, 0.03327, -0.04731, 0.16218, -0.57010, 0.68145, -0.40511, 0.12839)

Corrected direction v = (0.35362, -0.68338, 0.56021, -0.28725, 0.10241, -0.03092, -0.00061, 0.01097, -0.00534)

Inputs: ar3_order_selection_warm_starts/processes/process_045.csv and replications/replication_045.csv (Padded warm start).

Calculation data: output/analysis/gradient_bound_byproduct/sixteen_failed_calculations.csv. Population reference: thesis Proposition prop:minimum_width.